Practice workbook · Express a part as a percentage of a total

One subgroup or component is expressed as a percentage of its whole, sometimes after totalling a table or to support a comparison between groups.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-PERCENTAGE-OF-TOTAL

1. Question 7 · part b.ii · 2 marks

January 2023 Paper 1

Worked-example retry · Attempt part b.ii only.

7 People with polycystic kidney disease have cysts on their kidneys.

Cysts are sacs of fluid and they affect the functioning of the kidney.

Polycystic kidney disease is caused by a mutation of a gene.

(a) Explain how a gene mutation can affect the phenotype of an individual.

(3)

(b) Polycystic kidney disease can cause kidney failure.

Kidney failure can be treated using dialysis or a transplant.

(i) Describe the benefits of a transplant rather than dialysis as a treatment for kidney failure.

(2)

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1. Question 7 · part b.ii · 2 marks

January 2023 Paper 1

Worked-example retry · Attempt part b.ii only.

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(ii) In one year in the United Kingdom there were 4733 organ donations.

Of these donations, 3233 were kidney donations.

Calculate the percentage of organ donations that are kidney donations.

(2)

percentage = ... %

(iii) The number of kidney donations is much higher than the number of other organ donations such as heart or liver.

Give two reasons for the higher number of kidney donations.

(2)

1 .....

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2 .....

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1. Question 7 · part b.ii · 2 marks · Working and answer

Working and answer

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2. Question 4 · part a.iv · 2 marks

June 2019 Paper 1R

Further practice · Attempt part a.iv only.

4 The blood is filtered by the kidneys.

The filtrate formed passes through the kidney tubules and is changed into urine.

(a) The table lists the masses of different substances in blood, filtrate and urine passing through the kidneys in one day.

	Mass of substance in g per day		
Substance	**in blood**	**in filtrate**	**in urine**
protein	8000	0	0
glucose	180	180	0
urea	53	53	25
uric acid	9	9	1

Use the information from the table to answer these questions.

(i) Which substance does not pass into the filtrate? **(1)**

- ☒ **A** protein
- ☒ **B** glucose
- ☒ **C** urea
- ☒ **D** uric acid

(ii) The molecules of this substance do not pass into the filtrate.

What is the reason for this? **(1)**

- ☒ **A** the molecules are too large
- ☒ **B** the molecules are reabsorbed
- ☒ **C** the molecules are a waste product
- ☒ **D** the concentration of the molecules in the blood is too low

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2. Question 4 · part a.iv · 2 marks

June 2019 Paper 1R

Further practice · Attempt part a.iv only.

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(iii) A person has diabetes but is not receiving treatment.

Explain why the mass of glucose per day in the urine would be different for this person.

(3)

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(iv) Calculate the mass of urea in the urine as a percentage of the mass of urea in the filtrate.

(2)

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percentage = ... %

2. Question 4 · part a.iv · 2 marks

June 2019 Paper 1R

Further practice · Attempt part a.iv only.

(b) Kidney failure can be treated by a kidney transplant or by dialysis.

Discuss the advantages and disadvantages of kidney transplants.

(5)

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(Total for Question 4 = 12 marks)

2. Question 4 · part a.iv · 2 marks · Working and answer

Working and answer

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3. Question 3 · part a.ii · 2 marks

June 2019 Paper 2R

Further practice · Attempt part a.ii only.

3 A woman who smokes when she is pregnant risks harming her developing fetus.

The table lists the birth masses of 310 children born to non-smoking mothers or to mothers who smoke.

	Birth masses of children in kg					
	Less than 2.4	**2.4 – 3.1**	**3.2 – 3.5**	**3.6 – 3.9**	**More than 3.9**	**Total number of births in sample**
Number of children born to non-smoking mothers	4	42	70	60	34	210
Number of children born to mothers who smoke	3	24	44	23	6	100

(a) (i) What is the most common birth mass in kg?

(1)

- ☒ **A** 2.4 – 3.1
- ☒ **B** 3.2 – 3.5
- ☒ **C** 3.6 – 3.9
- ☒ **D** More than 3.9

(ii) Calculate the percentage of children with a birth mass of 3.6 kg or more in the sample of non-smoking mothers.

(2)

percentage = ...

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3. Question 3 · part a.ii · 2 marks

June 2019 Paper 2R

Further practice · Attempt part a.ii only.

(b) Use data from the table to deduce what can be concluded about the effect of smoking on birth mass.

(2)

(c) Suggest why smoking affects birth mass.

(5)

(Total for Question 3 = 10 marks)

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3. Question 3 · part a.ii · 2 marks · Working and answer

Working and answer

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4. Question 4 · part b.iii · 2 marks

November 2020 Paper 1R

Further practice · Attempt part b.iii only.

4 (a) Kwashiorkor is a deficiency disease that results from the lack of a certain nutrient in the diet.

Use words from the box to complete the passage about kwashiorkor.

(2)

fingers	carbohydrate	bread	ankles	protein	fat

Kwashiorkor is a disease that is more common in children.

One symptom of kwashiorkor is swelling in the ..... .

The symptoms of kwashiorkor are corrected by adding more ..... to the diet.

(b) The table gives information about the number of cases of kwashiorkor recorded in a hospital over four years.

Year	Number of cases of kwashiorkor
2010	32
2011	21
2012	23
2013	19

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4. Question 4 · part b.iii · 2 marks

November 2020 Paper 1R

Further practice · Attempt part b.iii only.

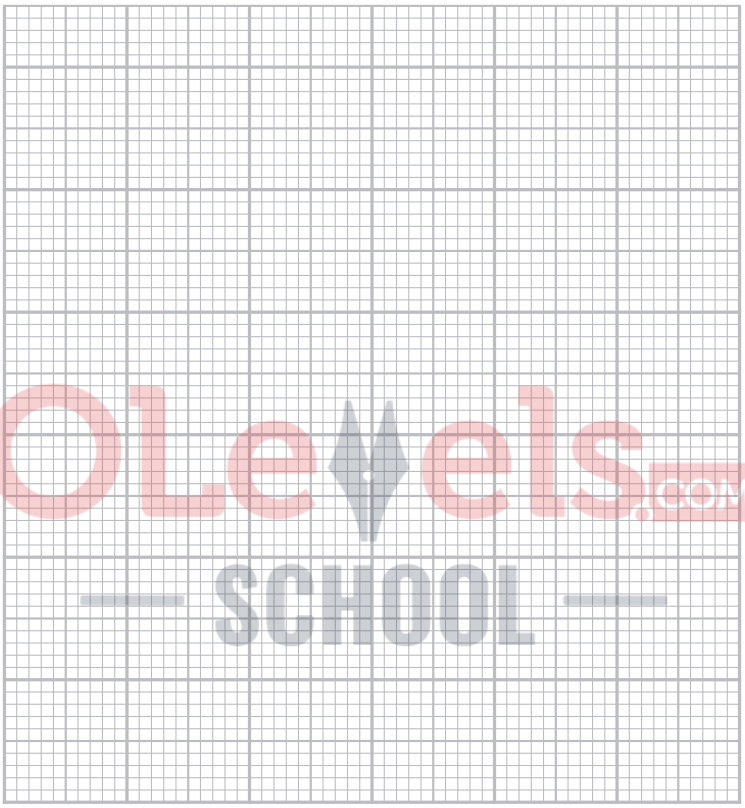
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(i) Plot a bar chart of the data on the grid.

(3)



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(ii) Suggest a reason for the difference in the number of recorded cases of kwashiorkor in 2010 compared with 2013.

(1)

.....

.....

(iii) In a hospital, 95 out of a total of 1978 children with malnutrition were found to have kwashiorkor.

Calculate the percentage of children with kwashiorkor.

(2)

percentage of children = ... %

(Total for Question 4 = 8 marks)

4. Question 4 · part b.iii · 2 marks · Working and answer

Working and answer

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5. Question 4 · part a.ii · 3 marks

November 2021 Paper 2

Further practice · Attempt part a.ii only.

4 The table shows the results of a survey about the number of teeth per student in a class of young students.

Number of teeth	Number of students
23	1
24	2
25	3
26	4
27	7
28	13

(a) (i) Draw a bar chart to show the results of the survey.

(4)



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5. Question 4 · part a.ii · 3 marks

November 2021 Paper 2

Further practice · Attempt part a.ii only.

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(ii) A full set of teeth for the students is 28

Calculate the percentage of the students in the class who have a full set of teeth.

(3)

percentage = ... %

(b) A full set of teeth for an adult is 32

Suggest why the full set of teeth for these students is only 28

(2)

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5. Question 4 · part a.ii · 3 marks · Working and answer

Working and answer

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6. Question 4 · part a.ii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part a.ii only.

4 (a) A student does a survey of eye colour in a sample of people.

The table shows a tally of the number of people with each eye colour in the sample.

Blue	Green	Brown	Hazel	Grey
𝍸 𝍸 𝍸 𝍸 𝍸	𝍸 𝍸 I	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 I	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸	III

(i) Complete the table by giving the missing information.

(1)

Number of people with each eye colour				
Blue	**Green**	**Brown**	**Hazel**	**Grey**
25				3

(ii) Calculate the percentage of people with brown eyes in the sample.

(2)

percentage = ... %

(b) Eye colour is determined by alleles inherited from parents.

(i) Which of these gives the definition of an allele?

(1)

- ☒ **A** two dominant genes in a genotype
- ☒ **B** two recessive genes in a genotype
- ☒ **C** different genes in a genotype
- ☒ **D** alternative forms of the same gene in a genotype

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6. Question 4 · part a.ii · 2 marks · Working and answer

Working and answer

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7. Question 8 · part b.i · 3 marks

June 2019 Paper 1

Further practice · Attempt part b.i only.

8 *P. vivax* is one of several parasites that can cause malaria.

(a) Explain how *P. vivax* is transmitted from one person to another.

(3)

(b) The table shows the total number of deaths from malaria and the number of deaths caused by *P. vivax*.

It shows the data for five different regions of the world in 2015.

Region of the world	Q	R	S	T	U
Total number of deaths from malaria	191 000	800	3800	14 400	1200
Number of deaths caused by *P. vivax*	1000	500	1400	4900	700

(i) Compare the number of deaths caused by *P. vivax* with the total number of deaths from malaria in regions Q and R.

Include calculations to support your answer.

(3)

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7. Question 8 · part b.i · 3 marks

June 2019 Paper 1

Further practice · Attempt part b.i only.

(ii) Suggest one reason why the number of deaths from malaria varies across the different regions.

(1)

(c) Scientists are developing a vaccine to protect against malaria.

Explain how vaccinating individual people will help to protect a whole population from malaria.

(3)

(d) *P. vivax* reproduces by asexual reproduction and sexual reproduction.

Explain why it is an advantage for a species to reproduce by both methods.

(3)

(Total for Question 8 = 13 marks)

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7. Question 8 · part b.i · 3 marks · Working and answer

Working and answer

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